Roll No. | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 09

B.Tech.(CE/ECE/Electronics & electrical Engg.) (2012 to 2017)
B.Tech. (EE) (2012 Onwards)/Electrical & Electronics Engg.
(2011 Onwards)/B.Tech. (Electronics & Telecom Engg.)
(Sem.–3)

ENGINEERING MATHEMATICS – III

Subject Code : BTAM-301
M.Code : 56071

Time : 3 Hrs.

Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt ANY FOUR questions.
3. SECTION-C contains THREE questions carrying TEN marks each and students have to attempt ANY TWO questions.

SECTION-A

1. **Solve the following :**

a) Evaluate, $\int \frac{e^z}{(z+1)^2} dz$ along the circle $|z-3| = 3$.

b) Find, $L(t^2 e^{-2t} \cos t)$.

c) Form a partial differential equation from $z = f(x+ay) + g(x-ay)$.

d) Discuss the analyticity of $\sinh z$.

e) Define even and odd functions. Give an example of a function which is neither even nor odd.

f) What are Dirichlet's conditions for the expansion of $f(x)$ as a Fourier series in $(-\pi, \pi)$?

g) Find solution of homogeneous partial differential equation $r - s - 6t = 0$.

h) Find the bilinear transformation that map the points $z = 0, -i, -1$ into the points $w = i, 1, 0$.

i) Shows that $P_n(-x) = (-1)^n P_n(x)$.

j) State Cauchy's integral formula.

BR paper
exams preparatory

SECTION-B

2. Expand $f(x) = x \sin x$, $0 < x < 2\pi$ as a Fourier series.

3. Find $L^{-1}\left[\tan^{-1}\left(\frac{2}{s}\right)\right]$, where L^{-1} denotes inverse Laplace transform.

4. If $u+v=\frac{2\sin 2x}{e^{2y}+e^{-2y}-2\cos 2x}$ and $f(z) = u + iv$ is an analytic function of $z = x + iy$, find $f(z)$ in terms of z.

5. Solve the equation by method of separation of variables.

$$3\frac{\partial u}{\partial x}+2\frac{\partial u}{\partial y}=0 \text{ given } u = 4e^{-x} \text{ when } y = 0.$$

6. Show that $\int P_n(x)dx = \frac{1}{2n+1}[P_{n+1}(x)-P_{n-1}(x)]$.

SECTION-C

7. A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $y = a \sin \frac{\pi x}{l}$ from which it is released at time $t = 0$. Show that the displacement of any point at a distance x from one end at the t is given by $y(x, t) = a \sin\frac{\pi x}{l} \cos \frac{\pi ct}{l}$.

8. Find the power series solution about the origin of the equation

$$(1 - x^2)\, y'' - 4xy' + 2y = 0$$

9. Use the concept of residues to evaluate

$$\int_0^{2\pi} \frac{d\theta}{(a - b\sin\theta)};\ a > |b|$$

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.